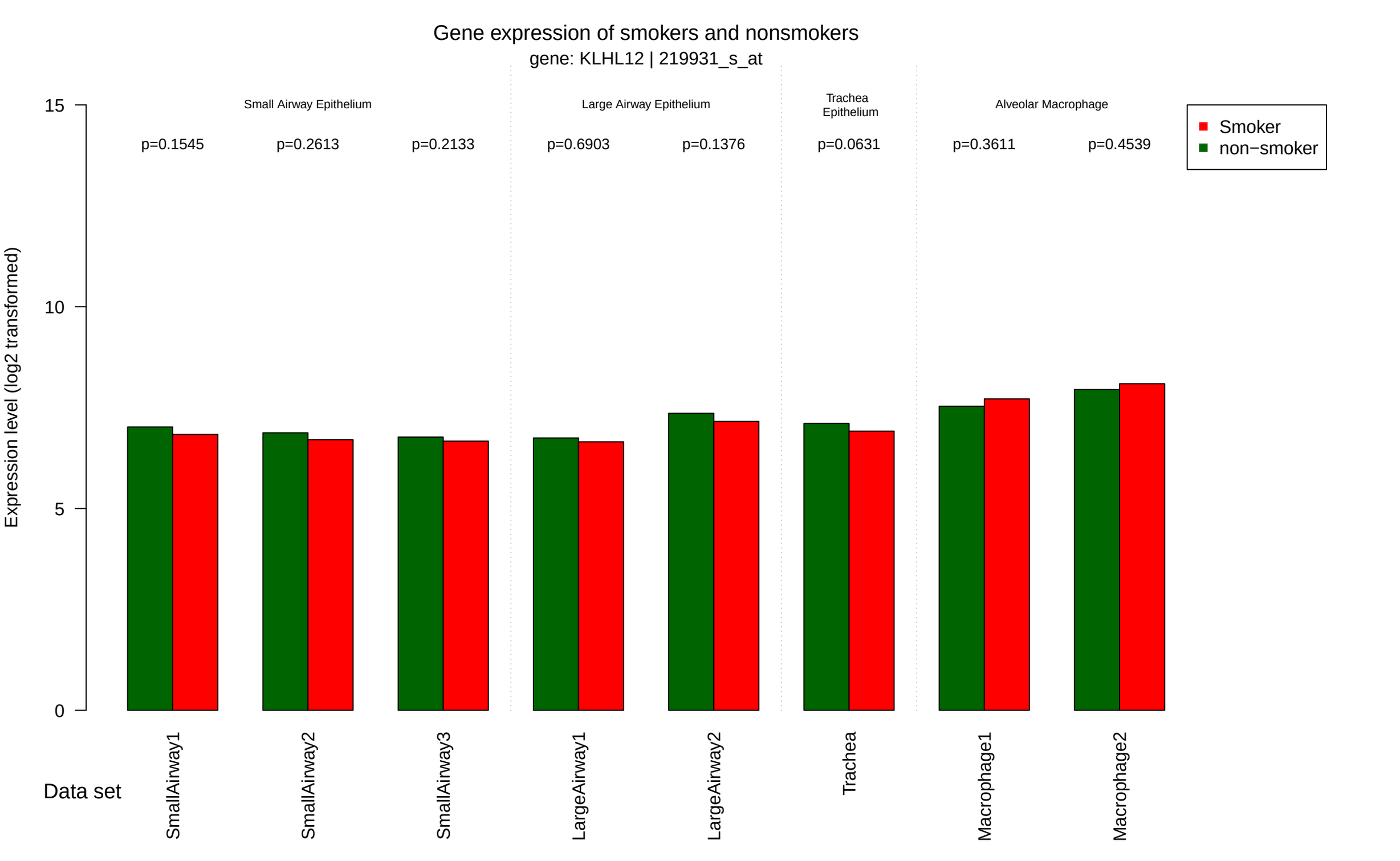
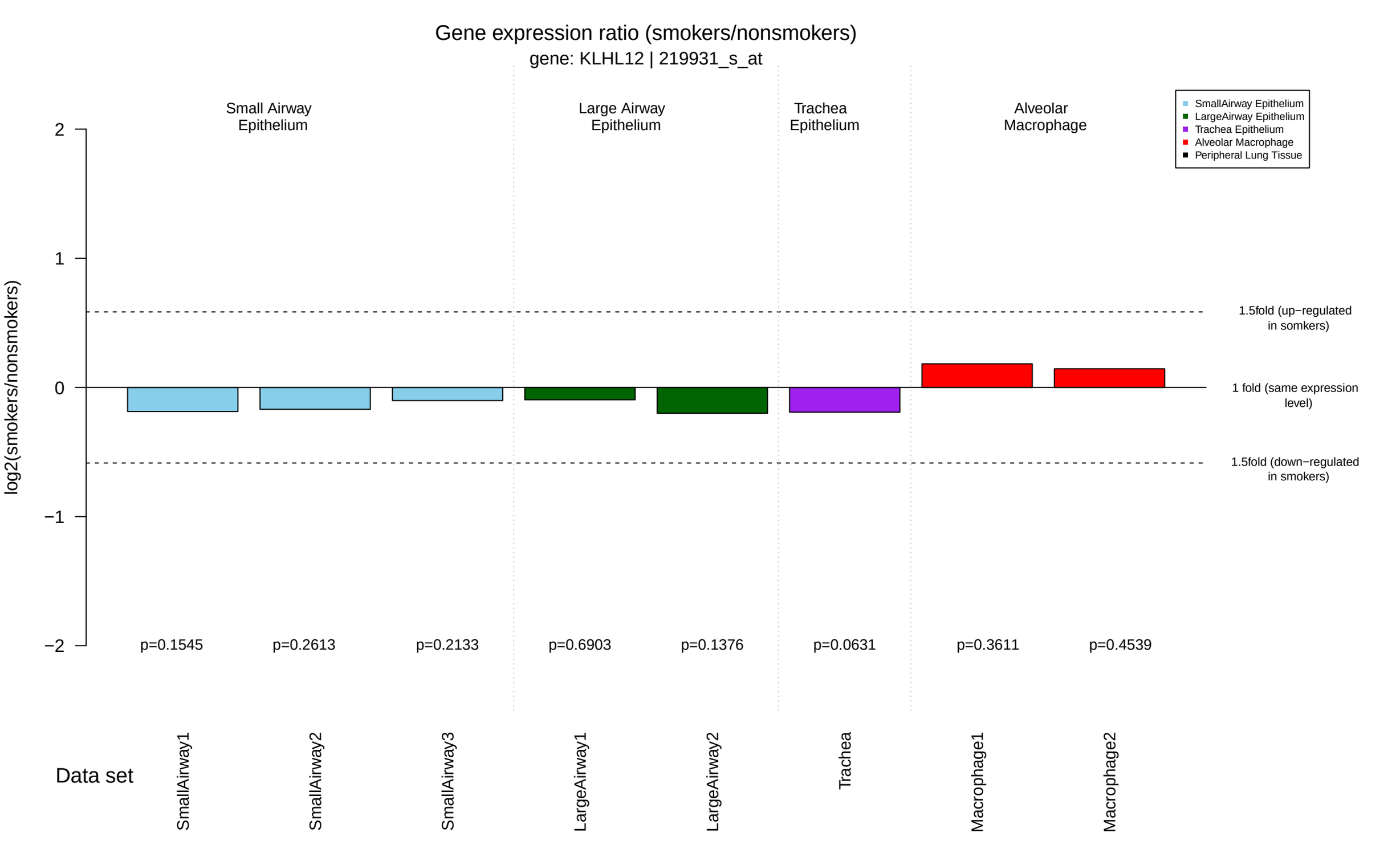
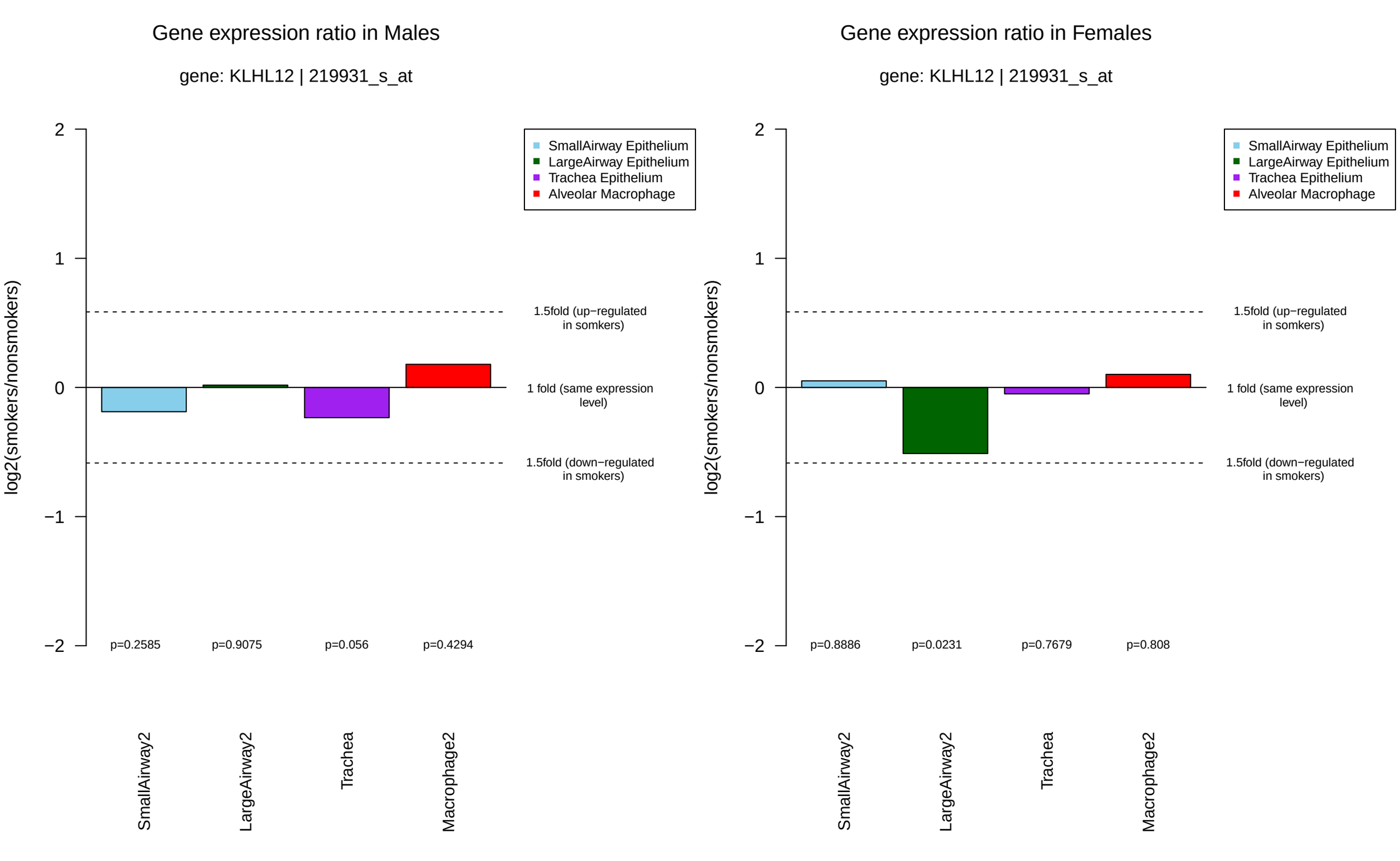
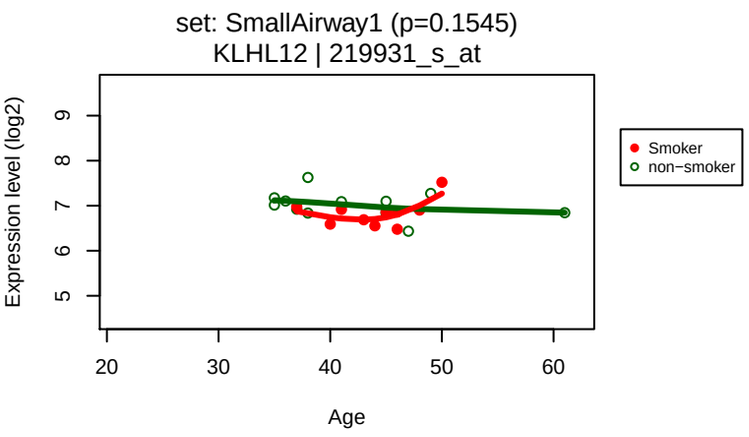
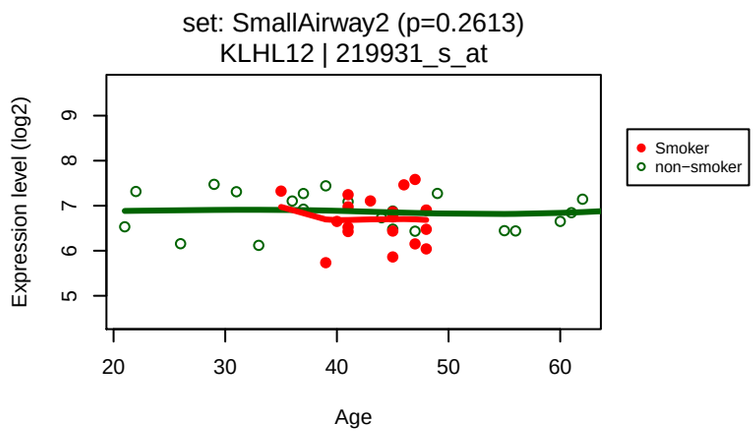
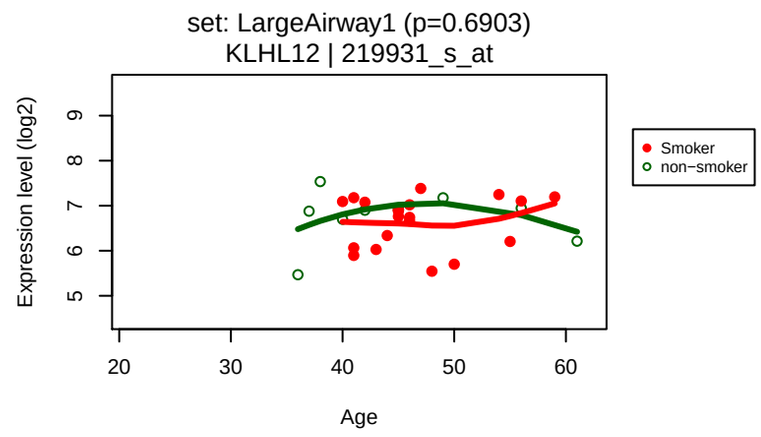

Gene expression of smokers and nonsmokers
gene: KLHL12 | 219931_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1545
p=0.2613
p=0.2133
p=0.6903
p=0.1376
p=0.0631
p=0.3611
p=0.4539
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

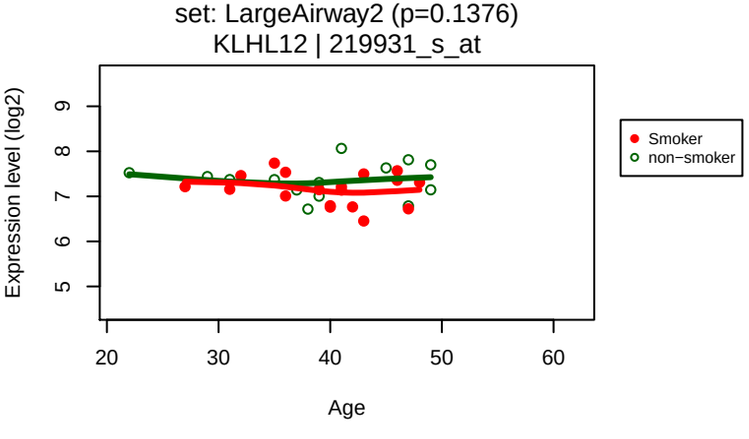

Gene expression ratio (smokers/nonsmokers)
gene: KLHL12 | 219931_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1545
p=0.2613
p=0.2133
p=0.6903
p=0.1376
p=0.0631
p=0.3611
p=0.4539
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

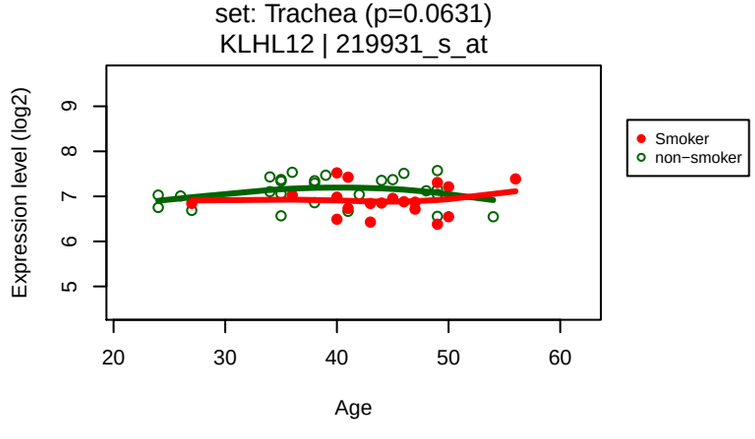

Gene expression ratio in Males
gene: KLHL12 | 219931_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2585
p=0.9075
p=0.056
p=0.4294
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KLHL12 | 219931_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8886
p=0.0231
p=0.7679
p=0.808
SmallAirway2
LargeAirway2
Trachea
Macrophage2

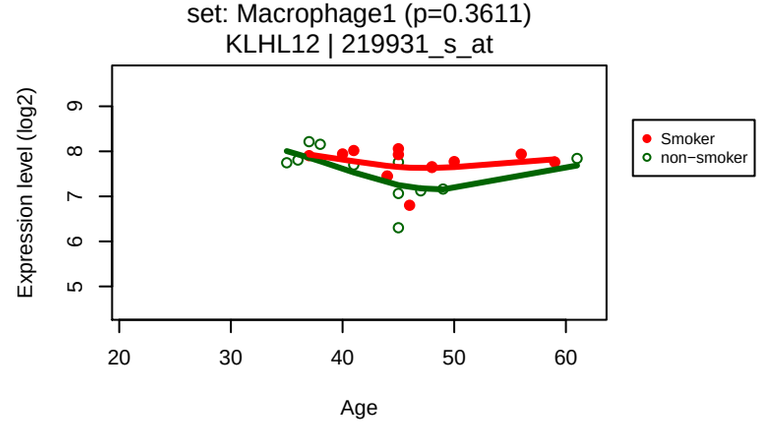

set: SmallAirway1 (p=0.1545)
KLHL12 | 219931_s_at
Expression level (log2)
Age
Smoker
non−smoker

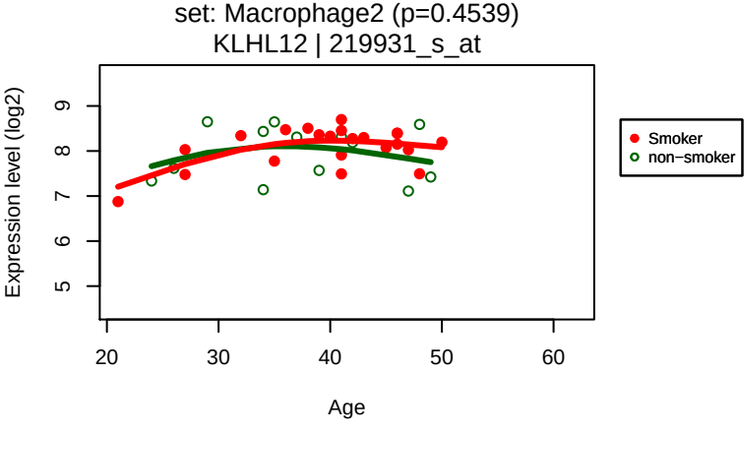

set: SmallAirway2 (p=0.2613)
KLHL12 | 219931_s_at
Expression level (log2)
Age
Smoker
non−smoker

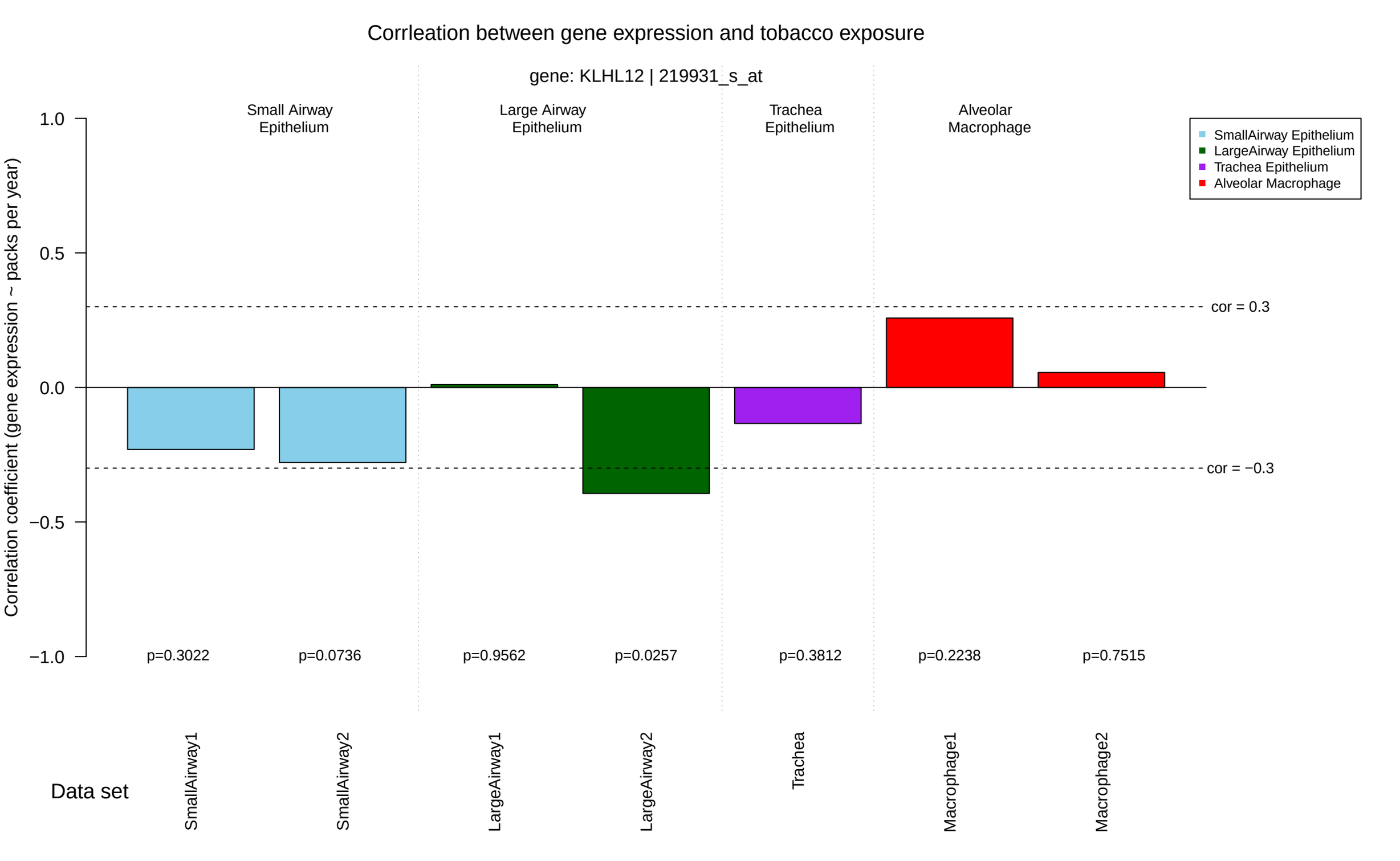

set: LargeAirway1 (p=0.6903)
KLHL12 | 219931_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1376)
KLHL12 | 219931_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0631)
KLHL12 | 219931_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3611)
KLHL12 | 219931_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4539)
KLHL12 | 219931_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KLHL12 | 219931_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3022
p=0.0736
p=0.9562
p=0.0257
p=0.3812
p=0.2238
p=0.7515
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2